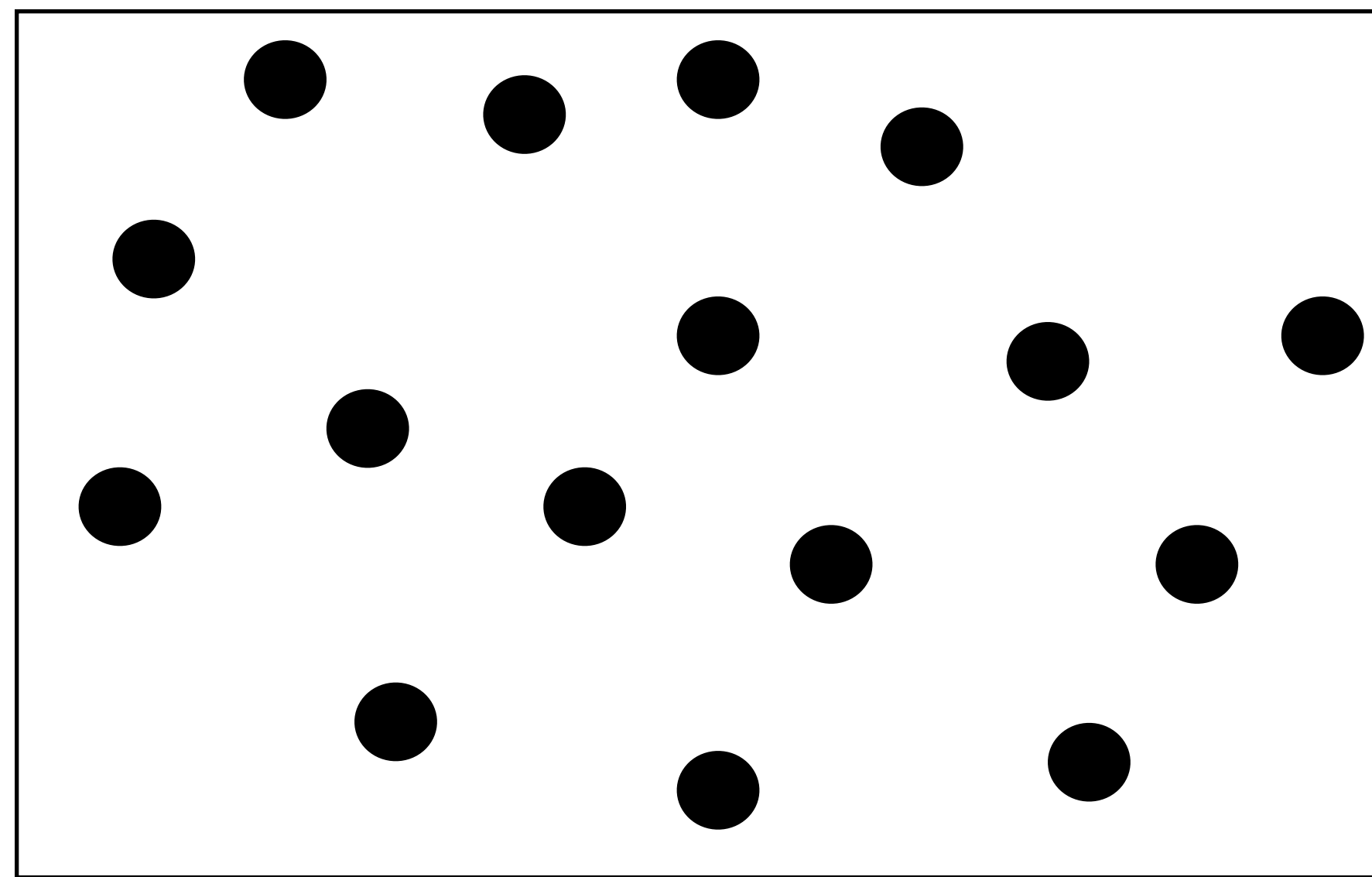


Towards Fair Representation: Clustering and Consensus

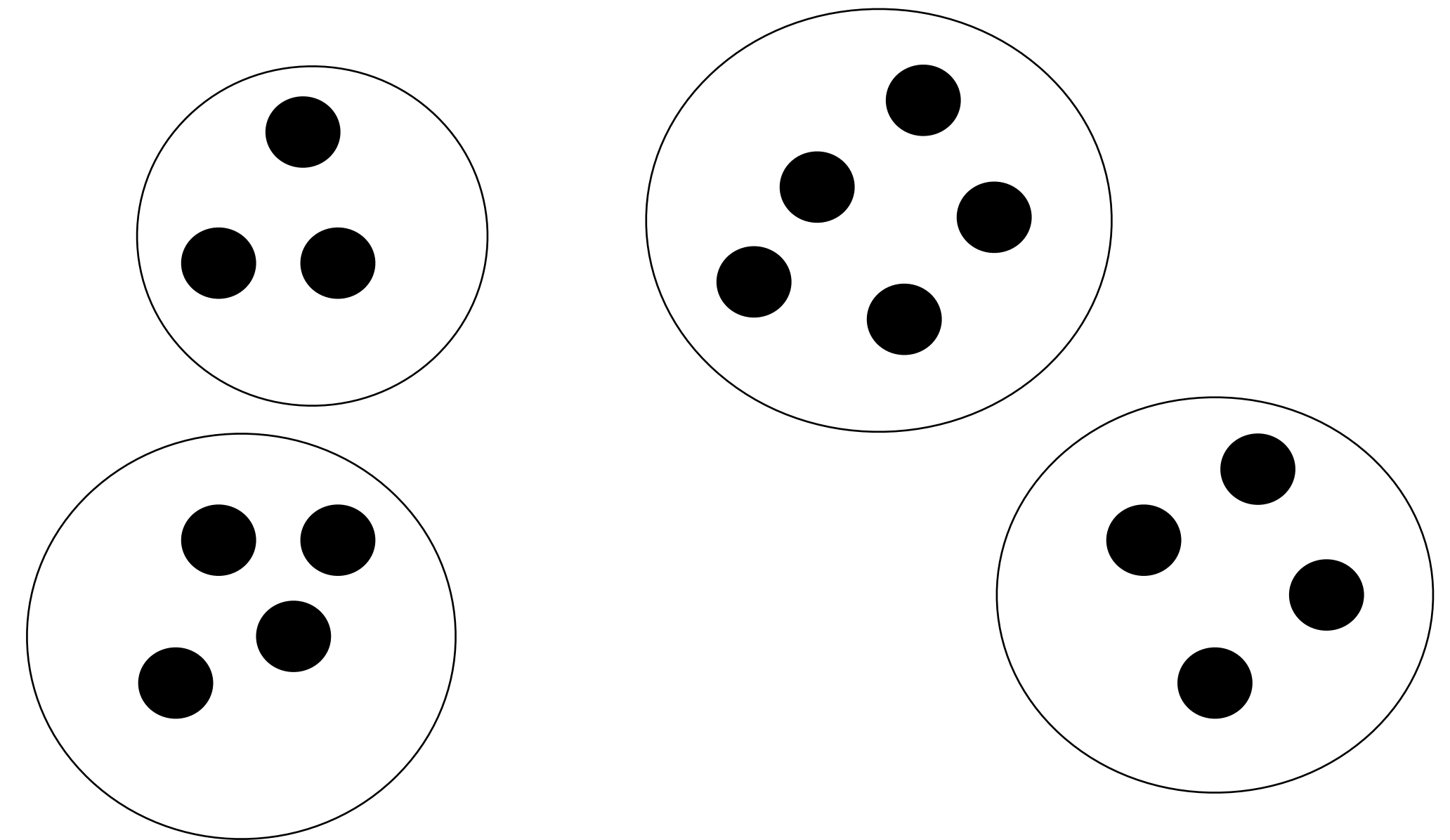
Diptarka Chakraborty, **Kushagra Chatterjee**, Debarati Das, Tien Long Nguyen and Romina Nobahari

What is Clustering?

- Given a set of data points, clustering is partitioning of data points based on some criteria.



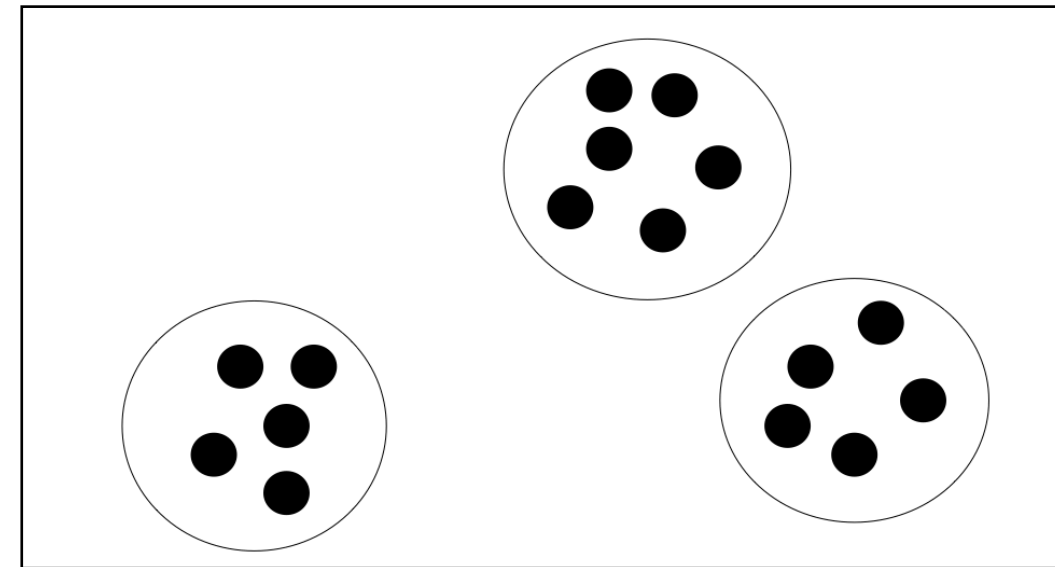
Students



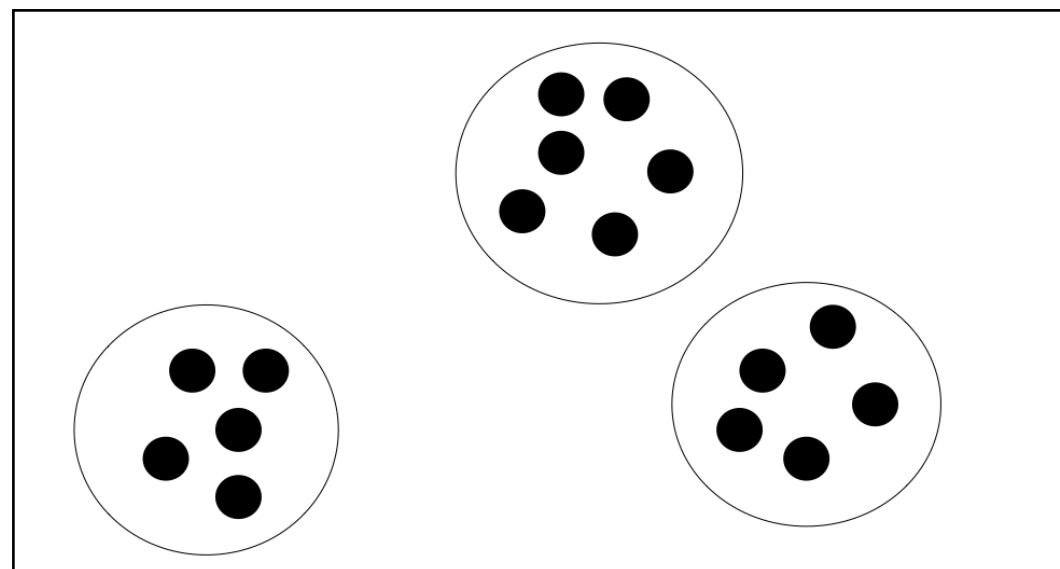
Clustering based on
academic performance

Consensus Clustering (Motivation)

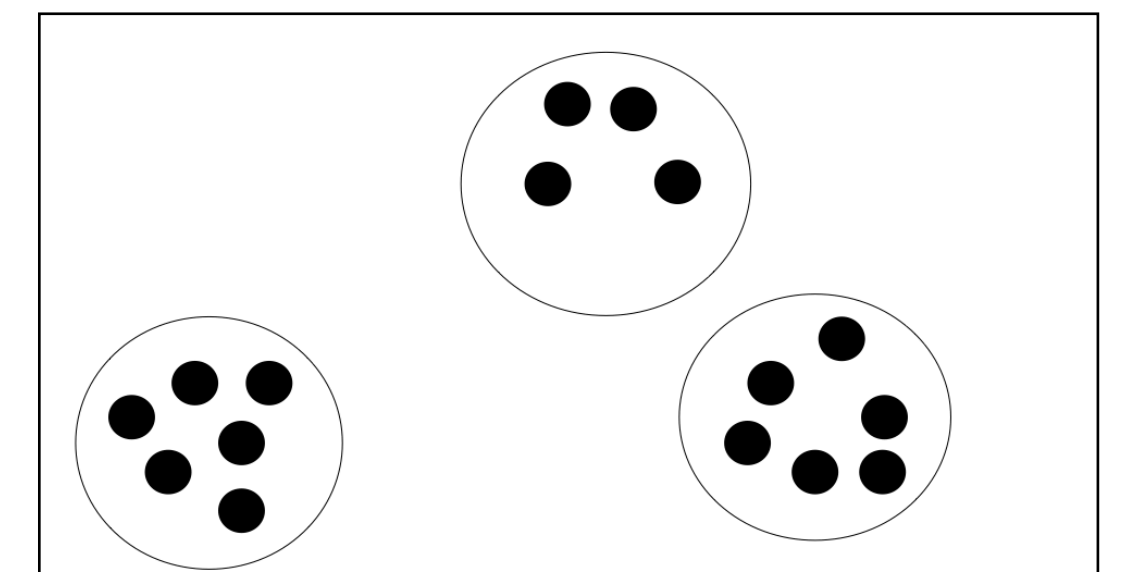
- Suppose a university wants to create a clustering of students to create **peer-support groups**.



Academic
Performance



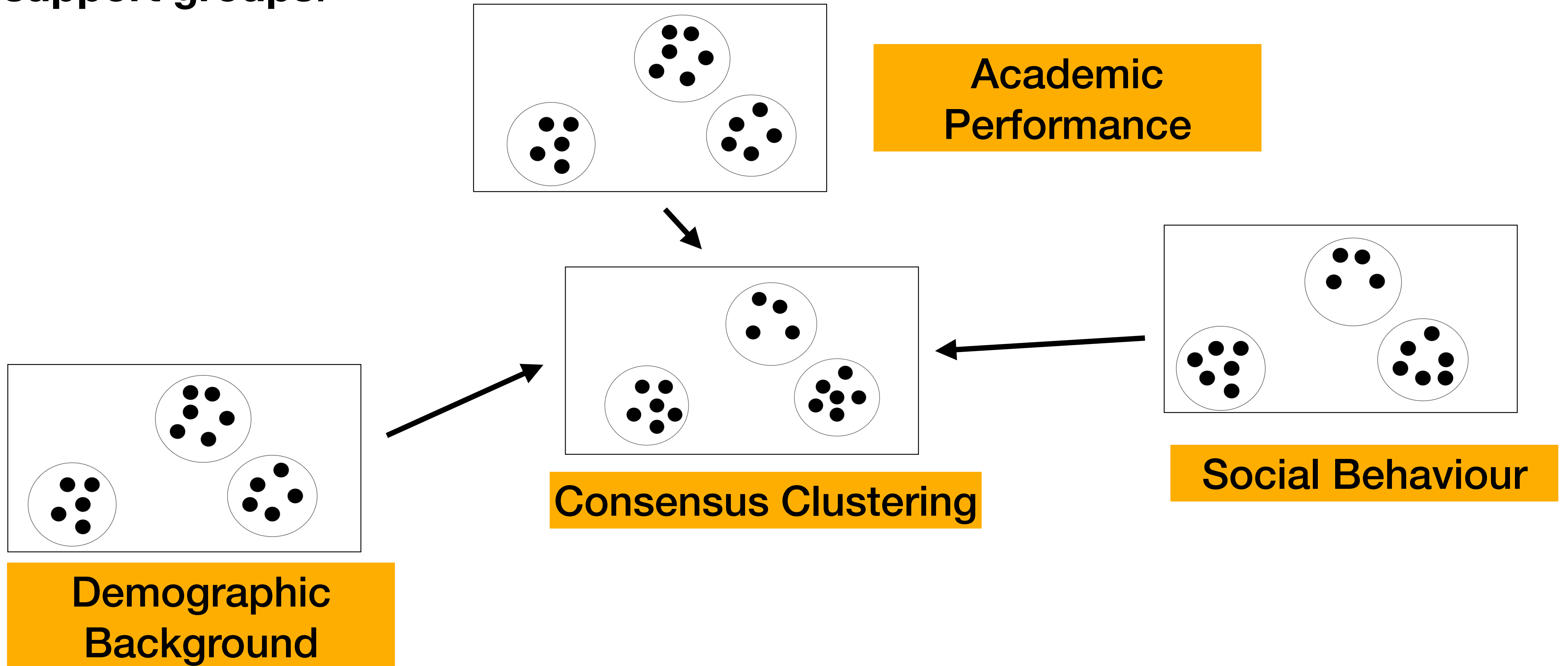
Demographic
Background



Social Behaviour

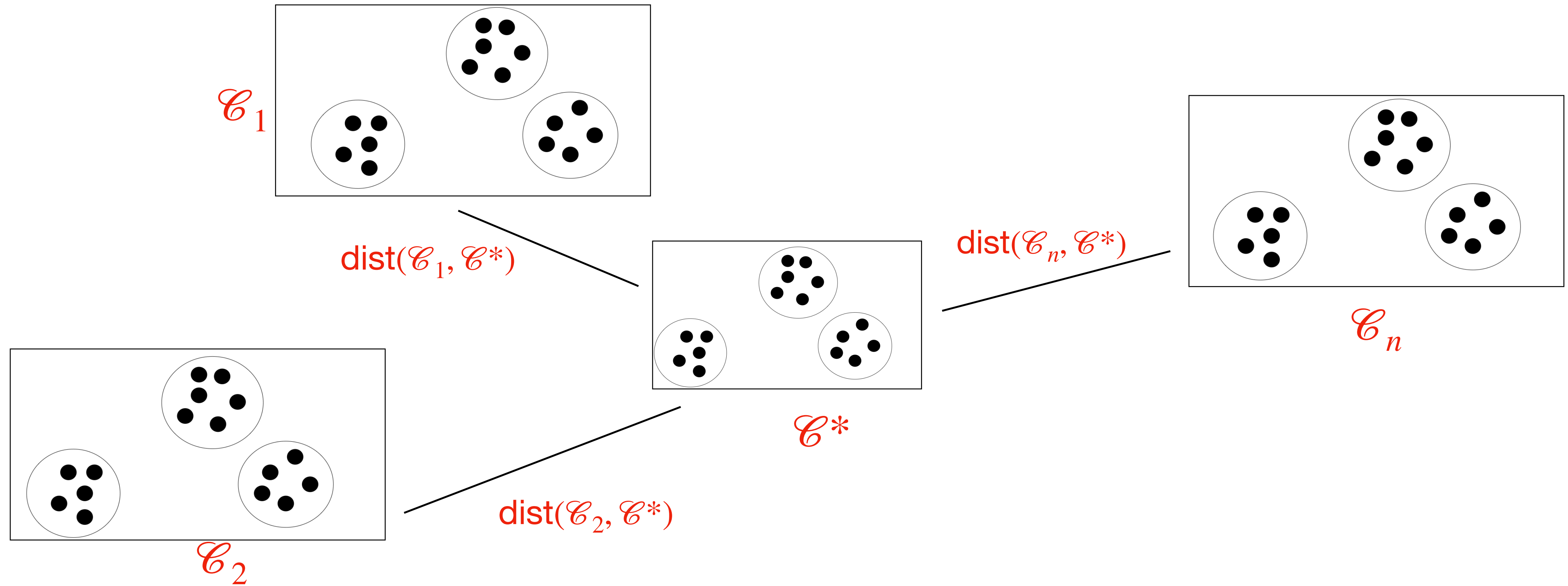
Consensus Clustering (Motivation)

- But the university wants to create one clustering of students to create **peer-support groups**.



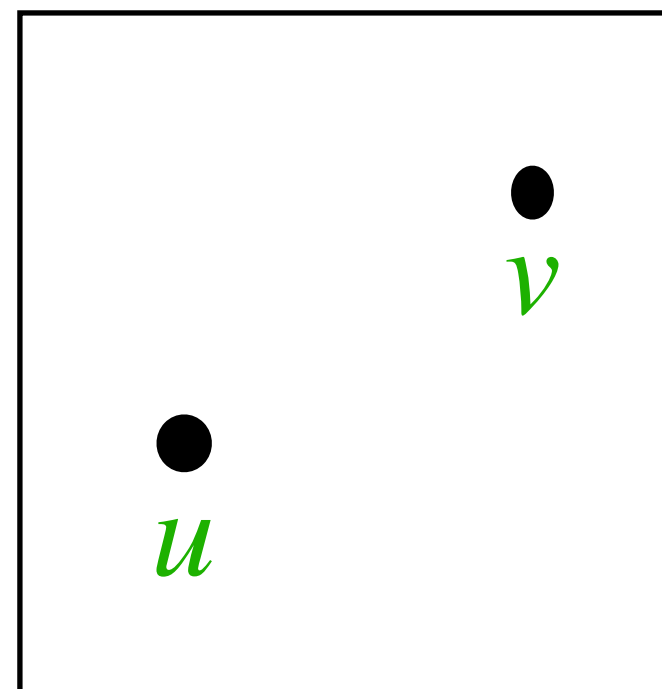
What is Consensus Clustering?

- Given clusterings $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_n, \mathcal{C}^*$ minimises the objective $(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell)^{1/\ell}$

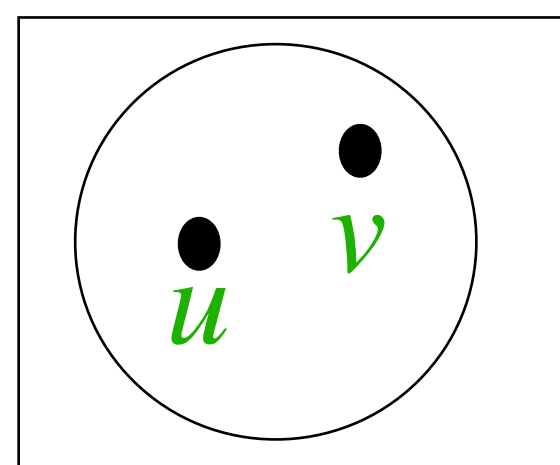


Distance between two clusterings

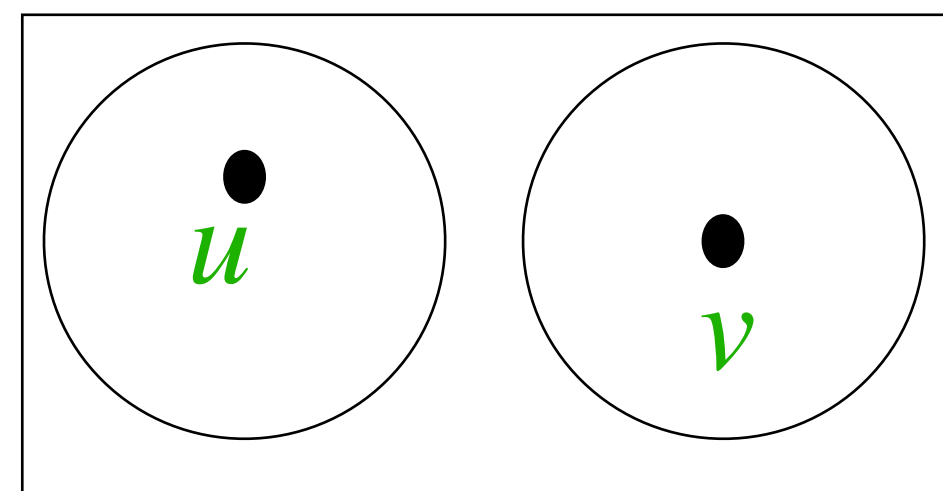
- $\text{dist}(\mathcal{D}, \mathcal{C})$: Number of pairs (u, v) that are together in one clustering but separated by other.



Suppose we are given two data points u, v and two clusterings $\mathcal{D}$ and $\mathcal{C}$ on these two points.



$\mathcal{D}$



$\mathcal{C}$

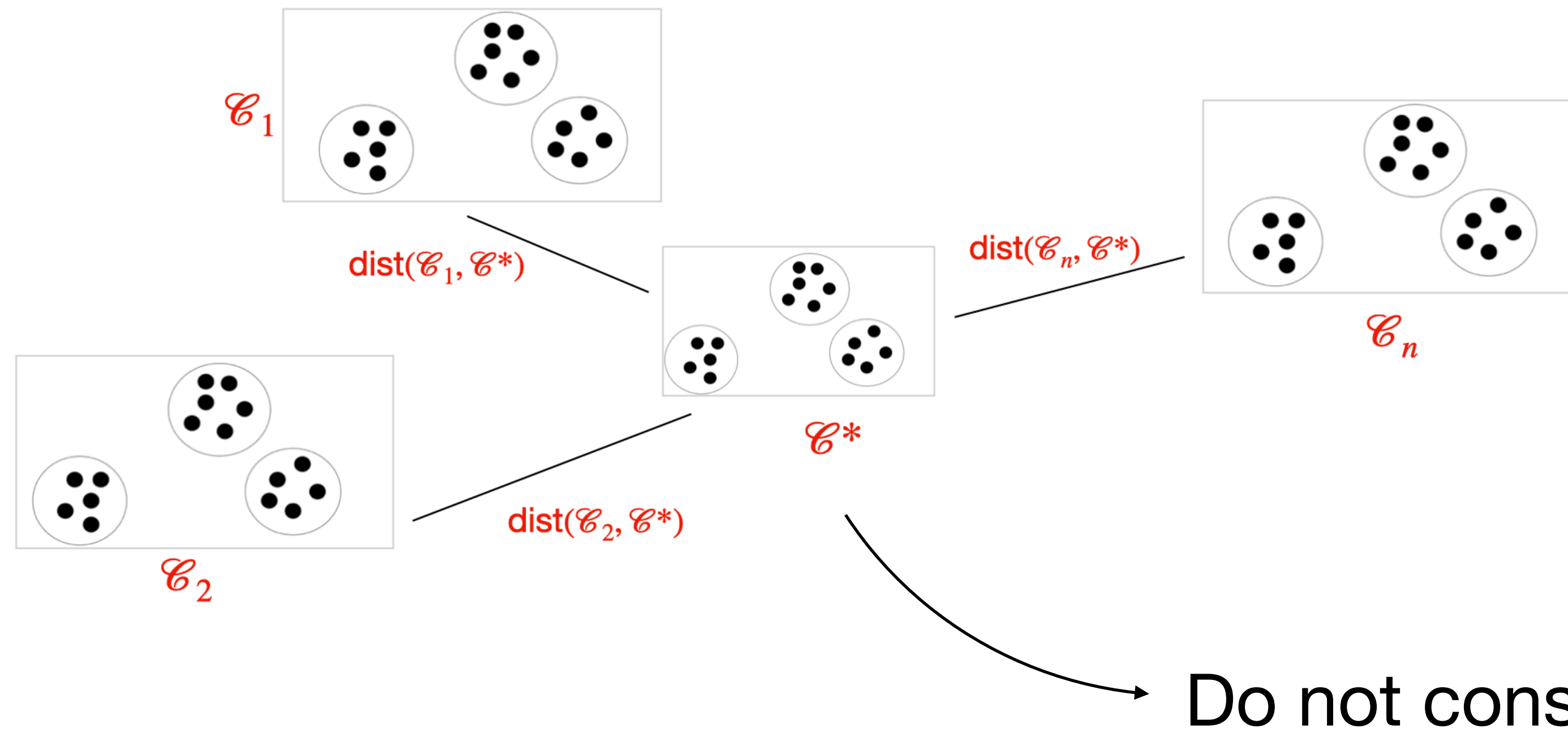
$$\text{dist}(\mathcal{D}, \mathcal{C}) = 1$$

Together in $\mathcal{D}$ but
separated in $\mathcal{C}$

Related Results of Consensus Clustering

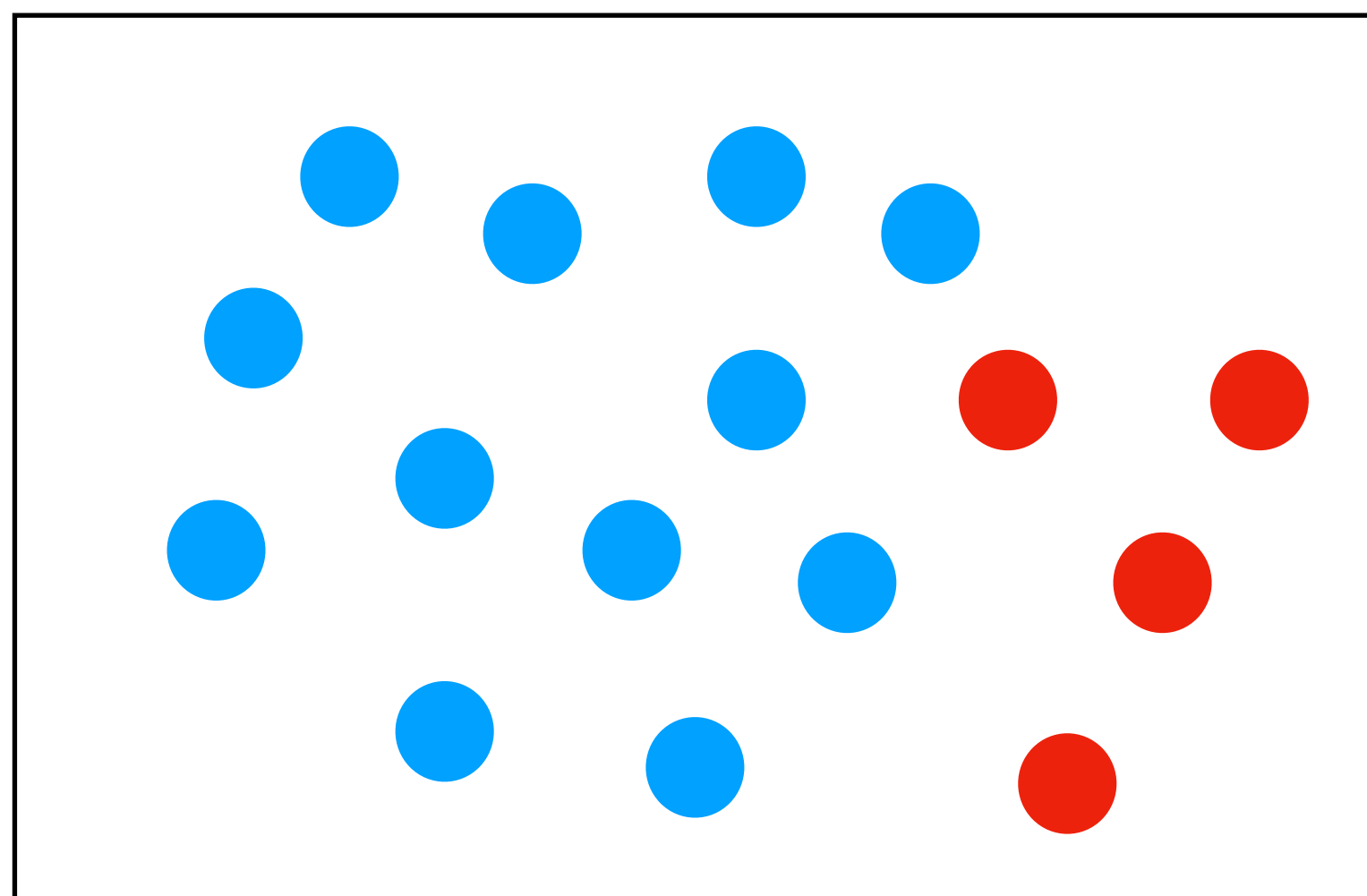
- Bonizzoni et al (JCSS 2008) : APX hard
- Folklore : **2 approximation**, Among the input clusterings $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_n$ return the clustering that minimises the objective $(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell)^{1/\ell}$
- Ailon, Charikar and Newman (JACM 2008) : **$11/7$ approximation** for **Median Consensus Clustering**, that is when $\ell = 1$ in the objective $(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell)^{1/\ell}$
- Das and Kumar (SODA 2025): **< 2 approximation** for **Centre Consensus Clustering**, that is when $\ell = \infty$ in the objective $(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell)^{1/\ell}$

Problem with Consensus Clustering



It may happen that some under represented minority students (lower-income students) are isolated in this clustering.

How is fairness represented?

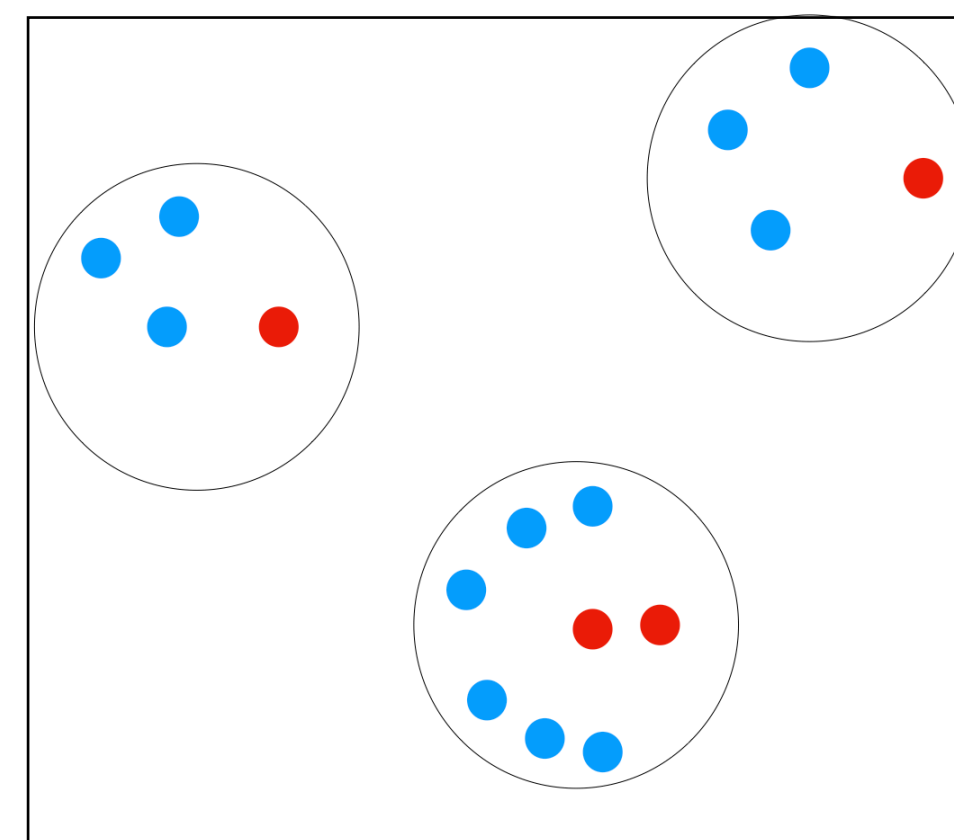


blue:red = $p : q$

Blue : Higher income students

Red : Lower income students

(Independent of the criteria used)



$\mathcal{F}$

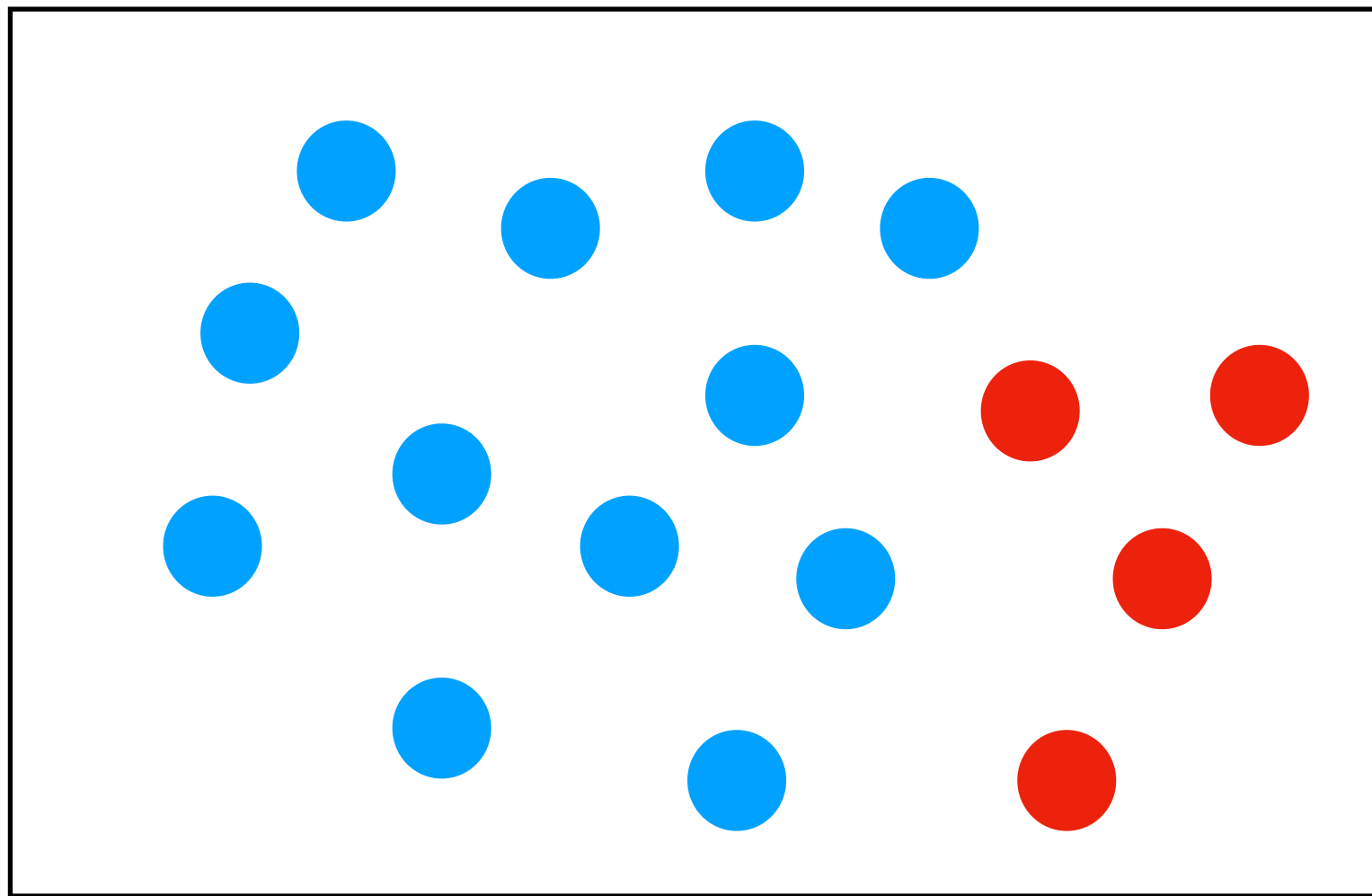
blue:red = $p:q$ is maintained for every cluster.

Introduced by Chierichetti et al (NeurIPS 2017)

Related Works of Fairness in Clustering

- Chierichetti et al (NeurIPS 2017): Introduced the notion of proportional fairness and used it for **k-median and k-centre clustering**.
- Ahmadian et al (AISTATS 2020, 2023), Ahmadi et al (ARXIV 2020): Studied proportional fairness for **Correlation clustering**.
- Used in **other contexts** such as **Rank Aggregation**, Celis et al (ICALP 2018), Chakraborty et al (NeurIPS 2022) and Kliachkin et al (ICDEW 2024).
- Our Work (COLT 2025): Studies proportional fairness for **Consensus Clustering**.

Problem Statement

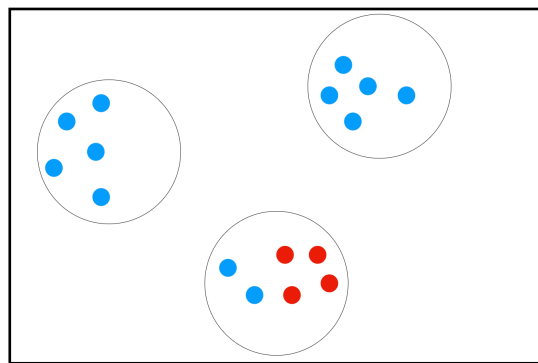


p, q s.t. blue:red = $p : q$

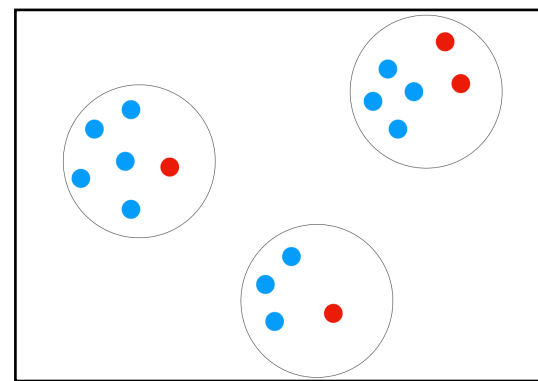
Two fairness parameters
given as input

+

$\mathcal{C}_1$

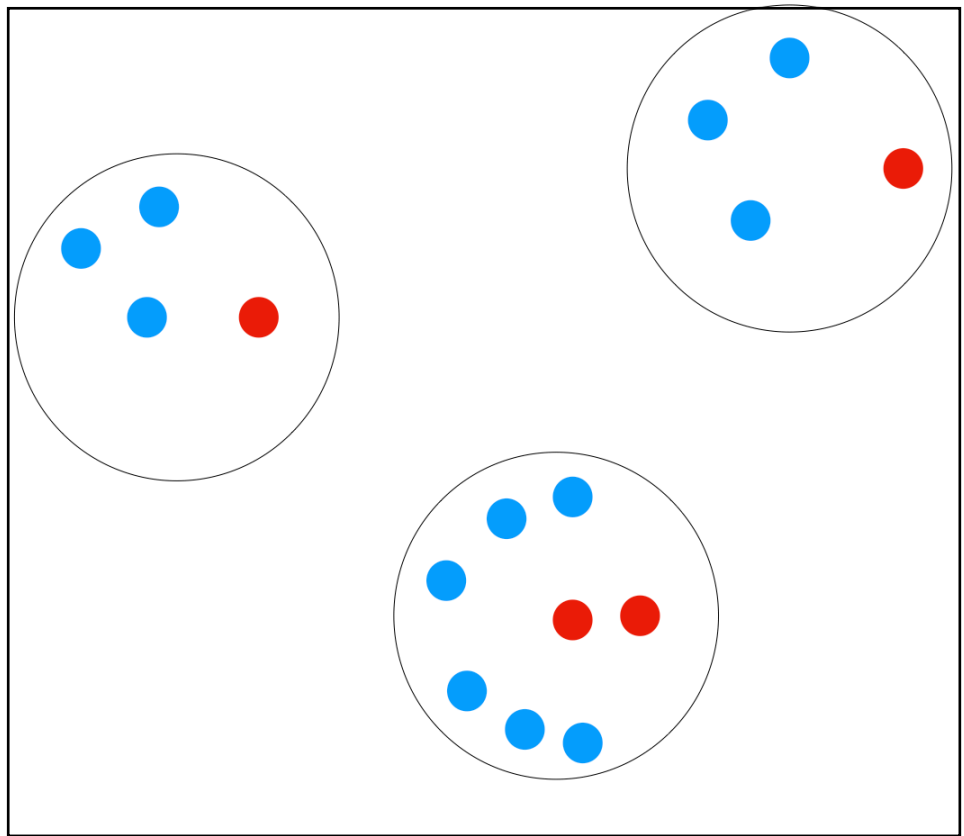
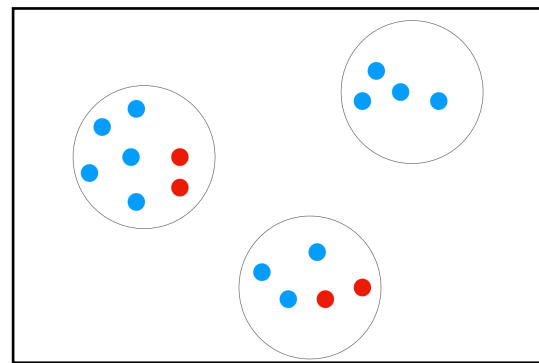


$\mathcal{C}_2$



⋮

$\mathcal{C}_n$



$\mathcal{F}^*$

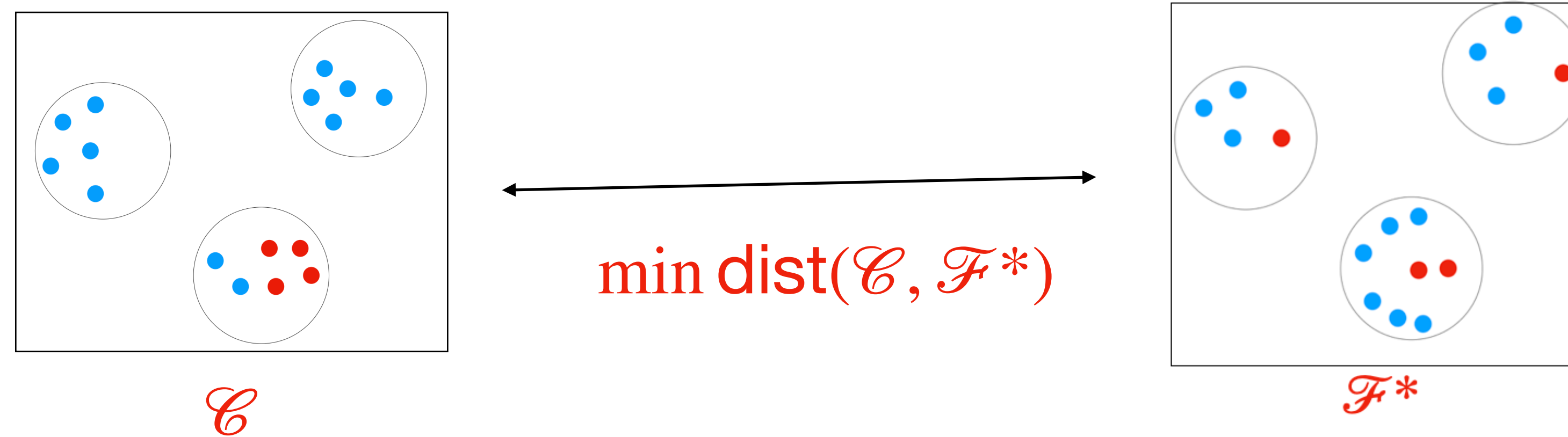
(Consensus Clustering s.t.
blue:red = $p:q$ in each cluster)

Our Contribution

Problem	Perfectly Balanced (1:1)	Integral Ratio (p:1)	Fractional Ratio (p:q)
Fair Consensus Clustering	3-approximation	19-approximation	35-approximation

How do we approach the problem?

- Suppose we are given only one clustering as input.



Goal : Find a fair consensus clustering for $\mathcal{C}$.

Closest Fair Clustering Problem

- Given an **unfair clustering**, can we find a **closest fair clustering** to it?



Closest Fair Clustering Problem

- Given an **unfair clustering**, can we find a **closest fair clustering** to it?

If yes then we can have a

3 approximation algorithm for Fair Consensus Clustering

Even for

* n input clusterings

* any values of **p:q**

* any values of ℓ in

$$\left(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell \right)^{1/\ell}$$

Closest Fair Clustering Problem

- Given an **unfair clustering**, can we find a **closest fair clustering** to it?

If yes then we can have a

3 approximation algorithm for Fair Consensus Clustering

If we get, **α -close** Fair clustering then we have **$(2 + \alpha)$**
Fair Consensus Clustering

Even for

* n input clusterings

* any values of **p:q**

* any values of **ℓ** in

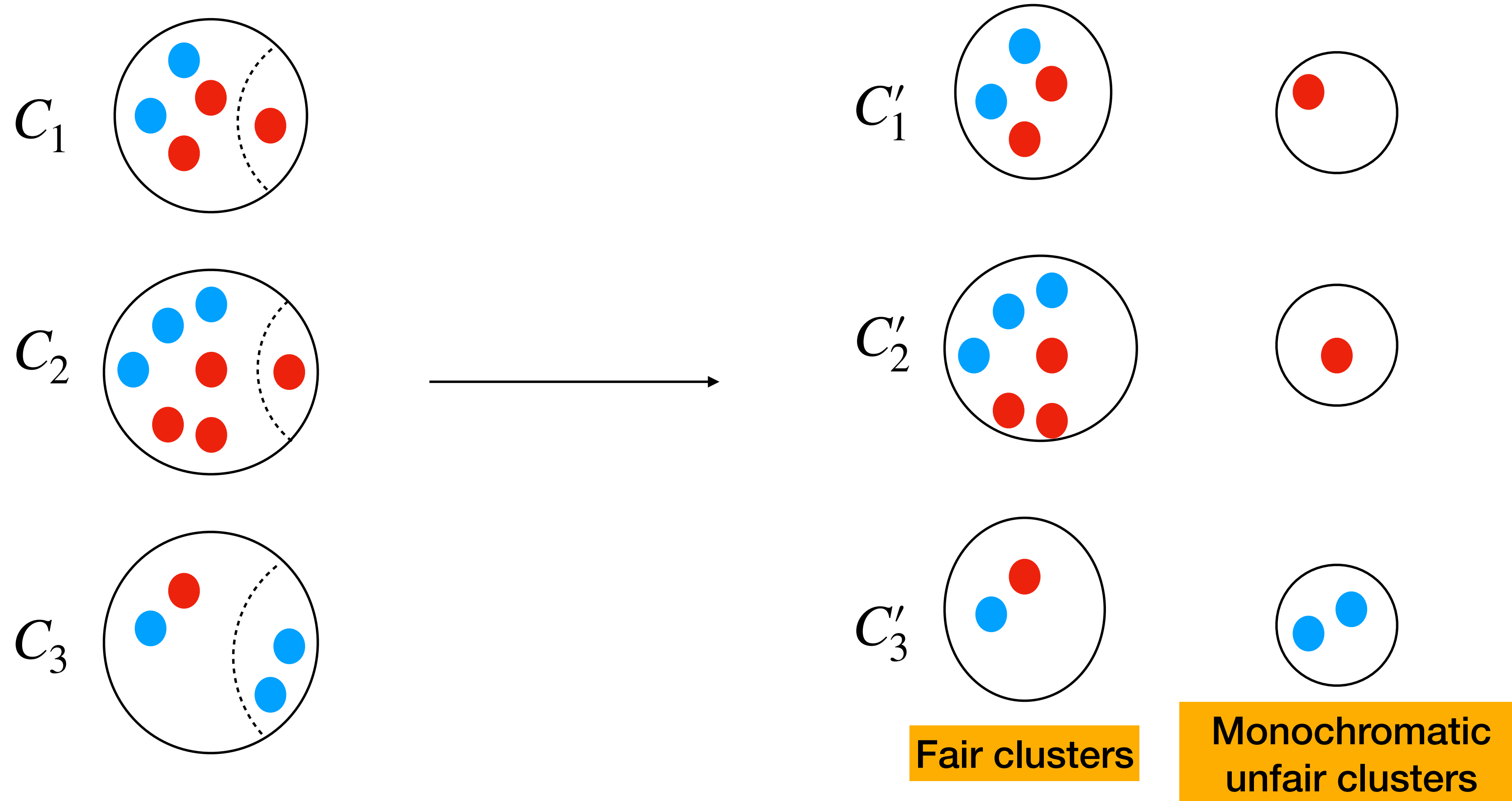
$$\left(\sum_{i=1}^n \text{dist}(\mathcal{C}_i, \mathcal{C})^\ell \right)^{1/\ell}$$

Our Contribution

- Closest Fair Clustering — NP-hard when $p \neq q$ (Reduction from 3-partition problem)
- Optimal algorithm for $1:1$ case and approximation algorithms for $p:1$ and $p:q$ case.

Problem	Perfectly balanced (1:1)	Integral Ratio (p:1)	Fractional Ratio (p:q)
Closest Fair Clustering	Optimal	17	33

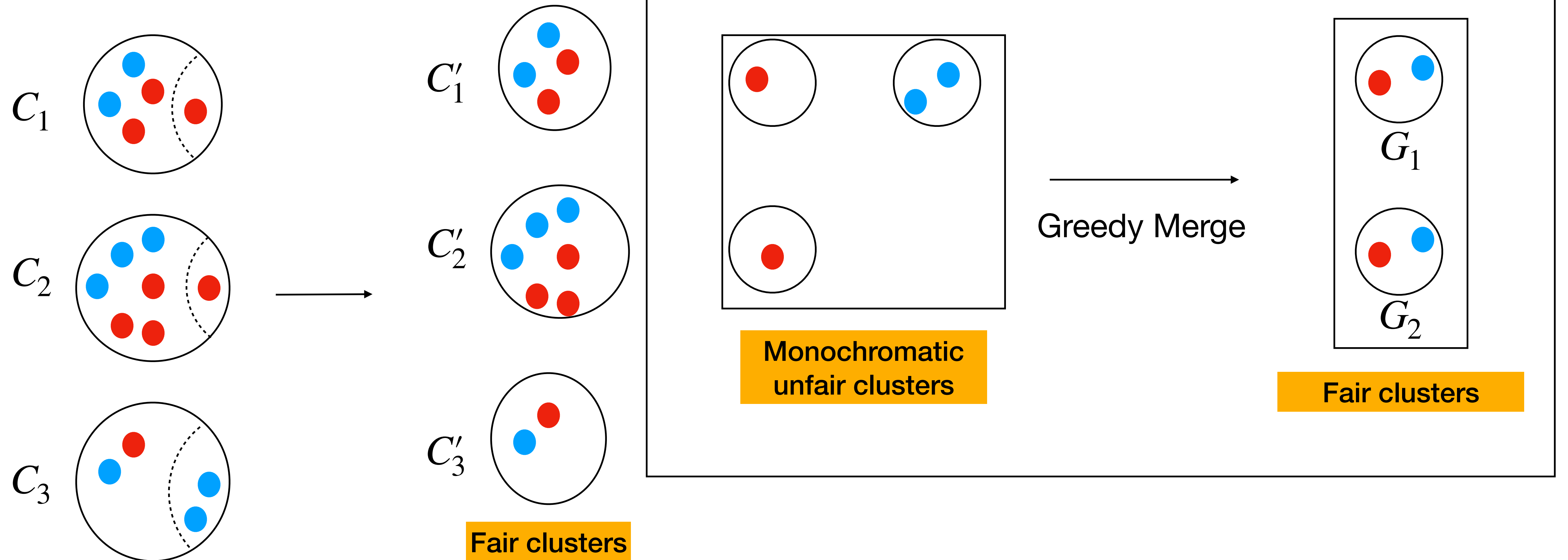
Algorithm for closest fair clustering (1:1)



Input Clustering $\mathcal{C}$

Cut the surplus part
from each cluster

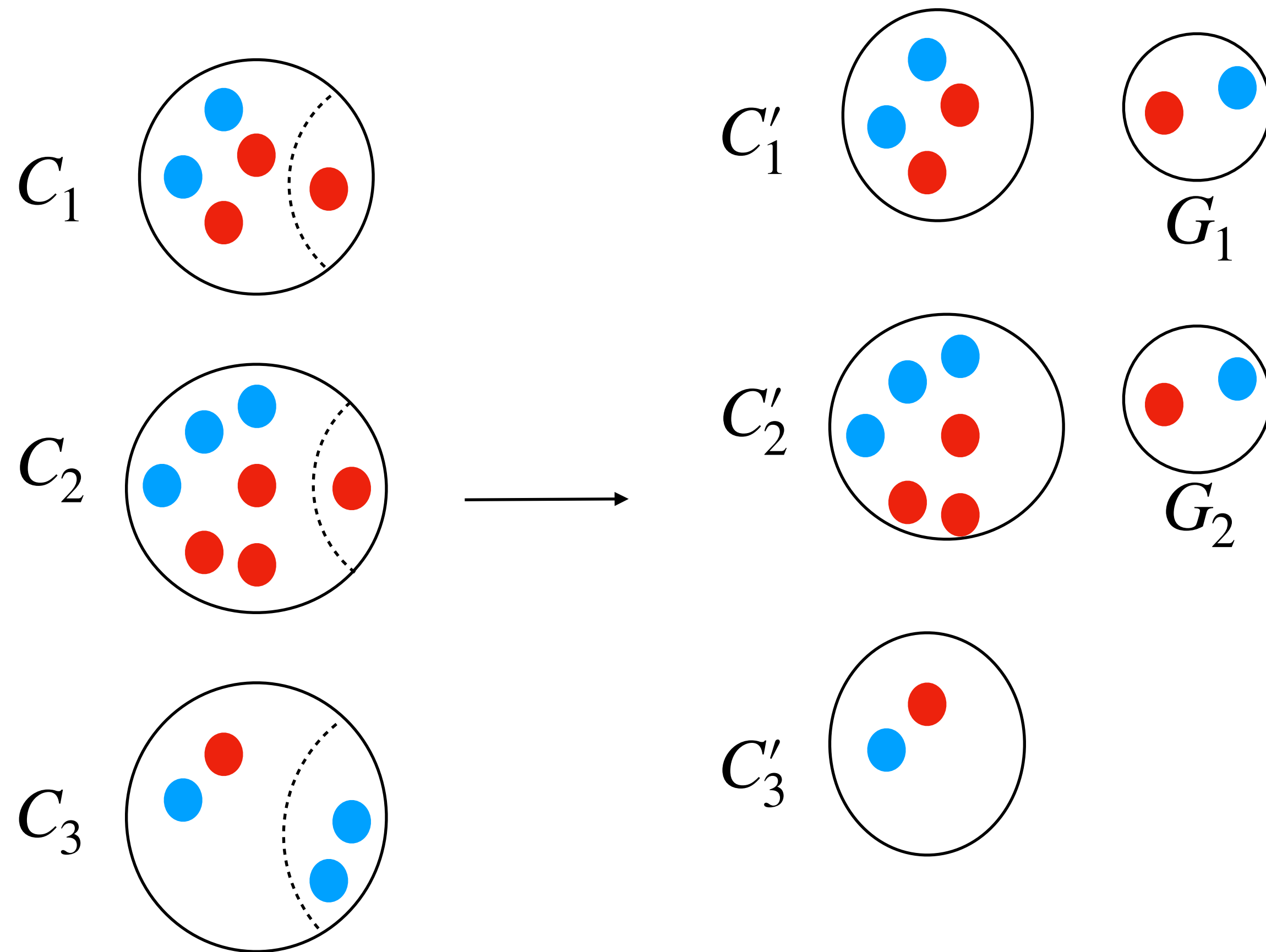
Algorithm for closest fair clustering (1:1)



Input Clustering $\mathcal{C}$

Cut the surplus part
from each cluster

Why our algorithm optimal?

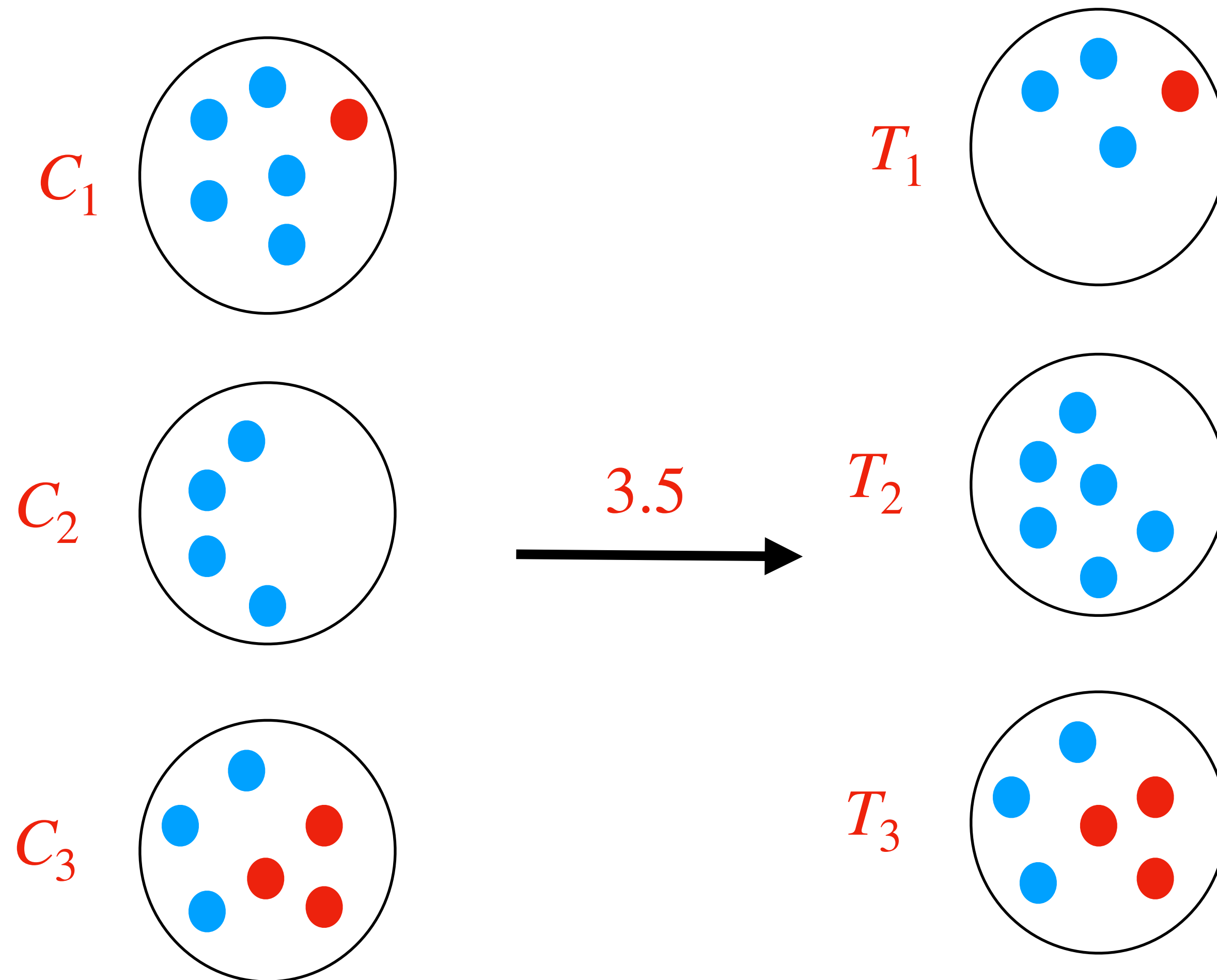


Input Clustering $\mathcal{C}$

Output- Fair clustering $\mathcal{F}$

We prove that the output $\mathcal{F}$ is one of the optimal solutions.

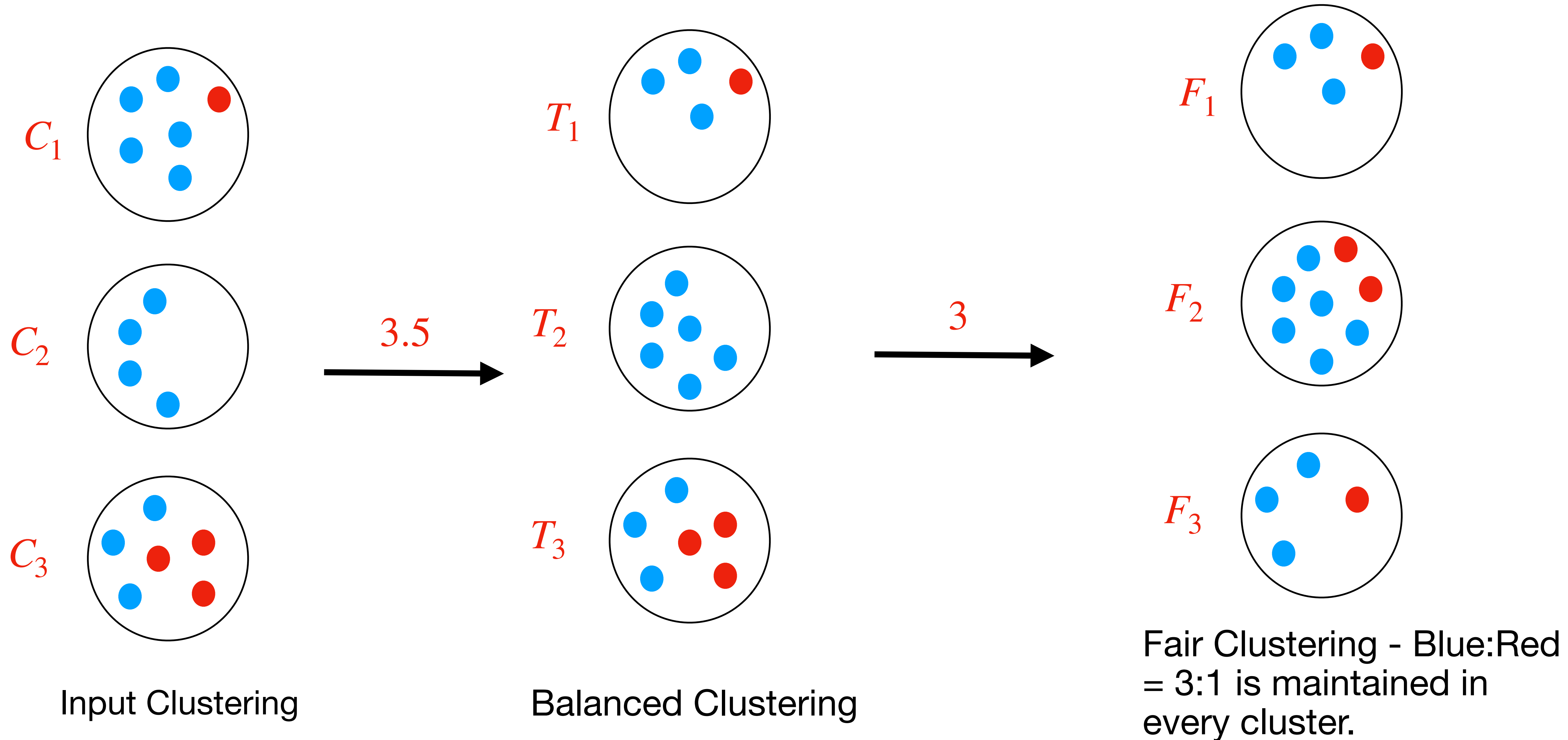
Approach for p:1 case



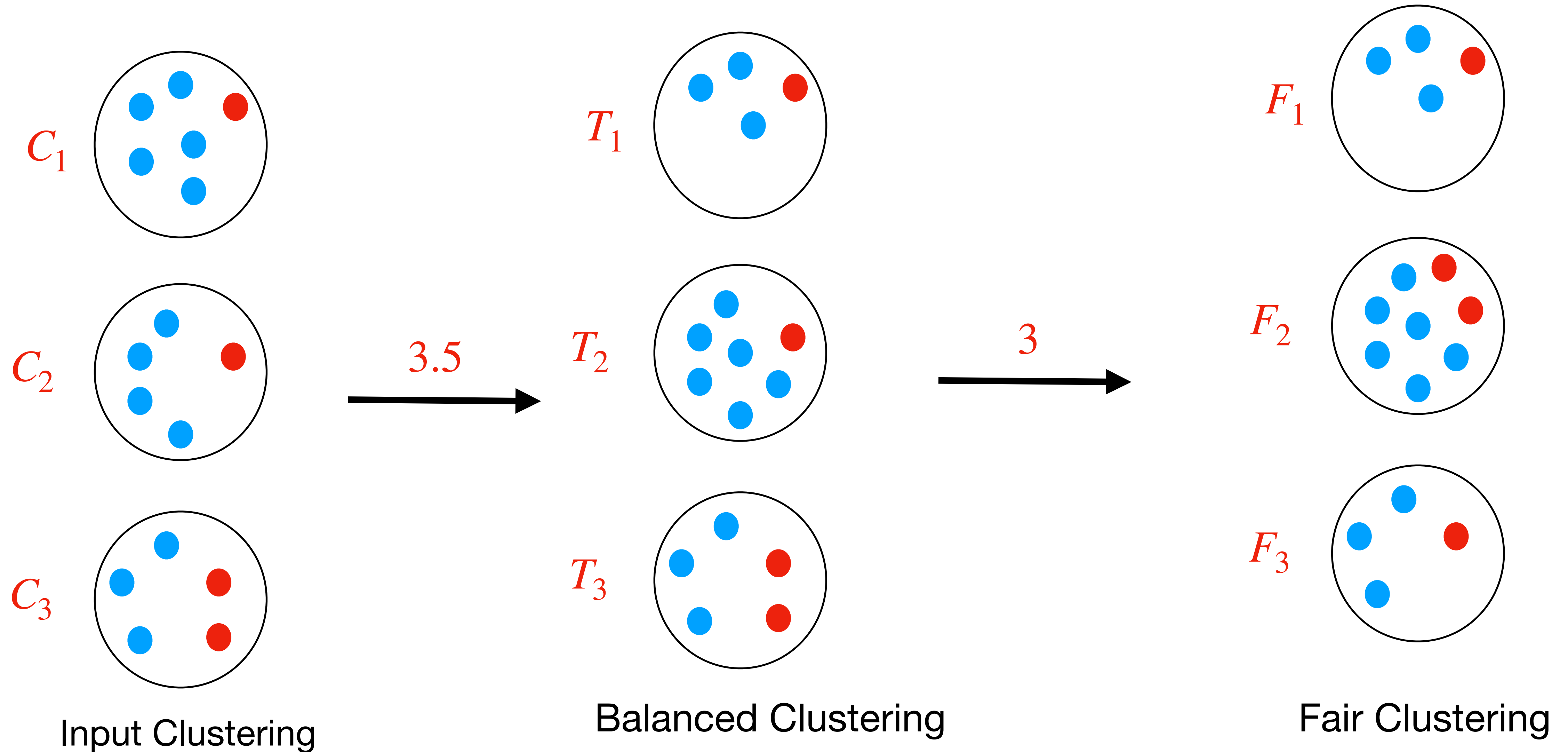
Input Clustering
(Blue:Red = 3:1 in
this example).

Balanced Clustering - In each cluster,
number of blue points is a multiple of
 p ($p = 3$, in this example).

Approach for p:1 case

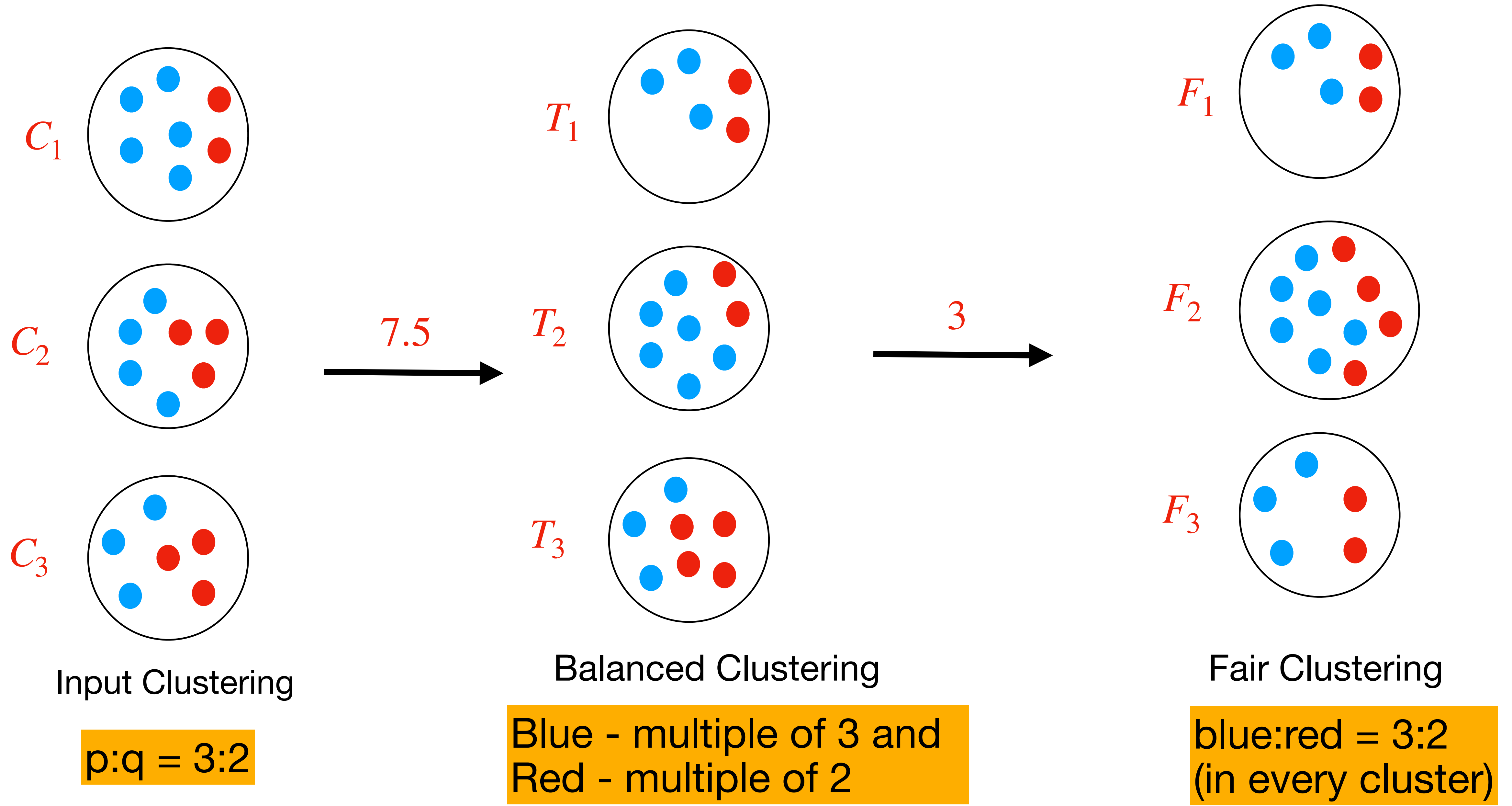


Approach for p:1 case

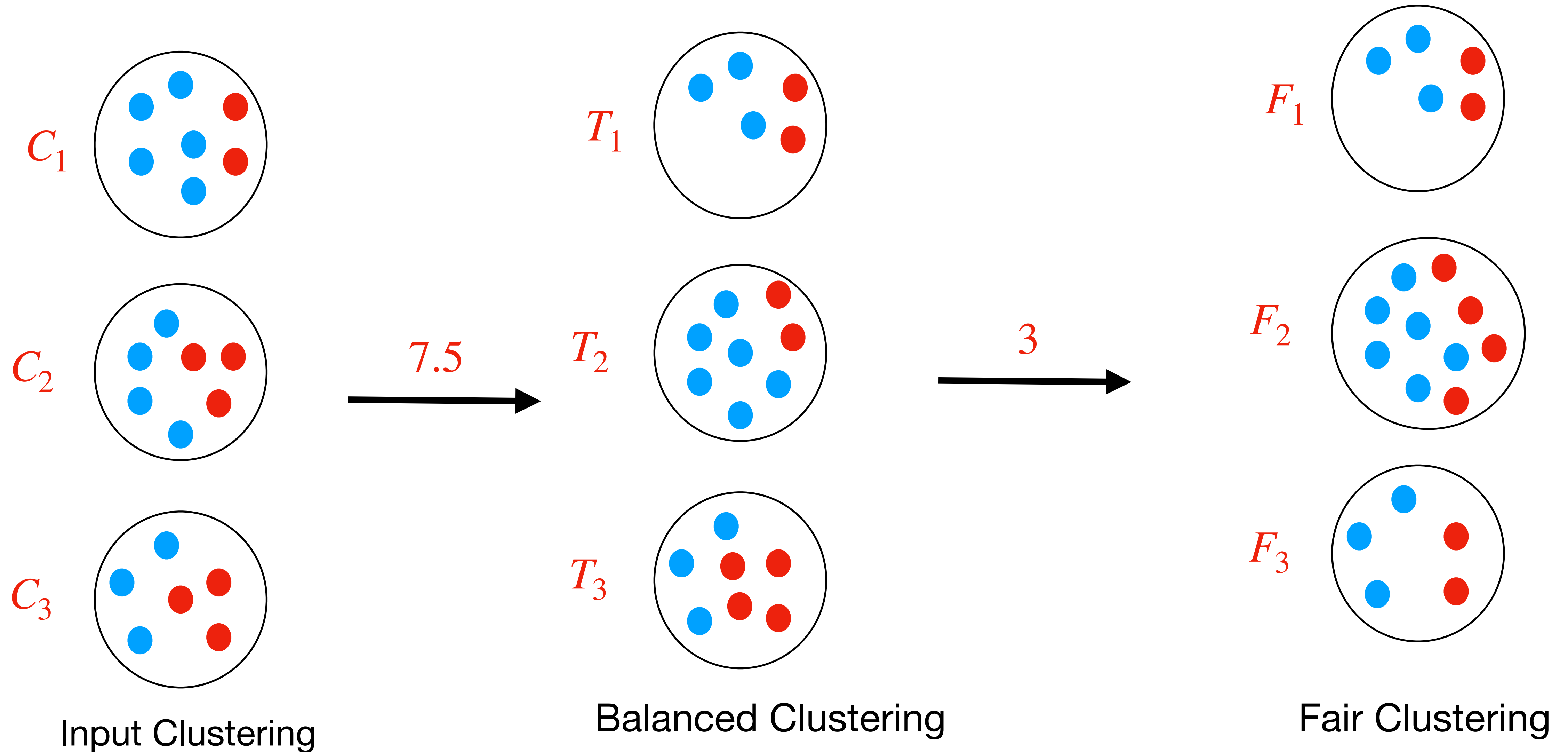


Overall approximation - 17 (independent of p)

Approach for p:q case

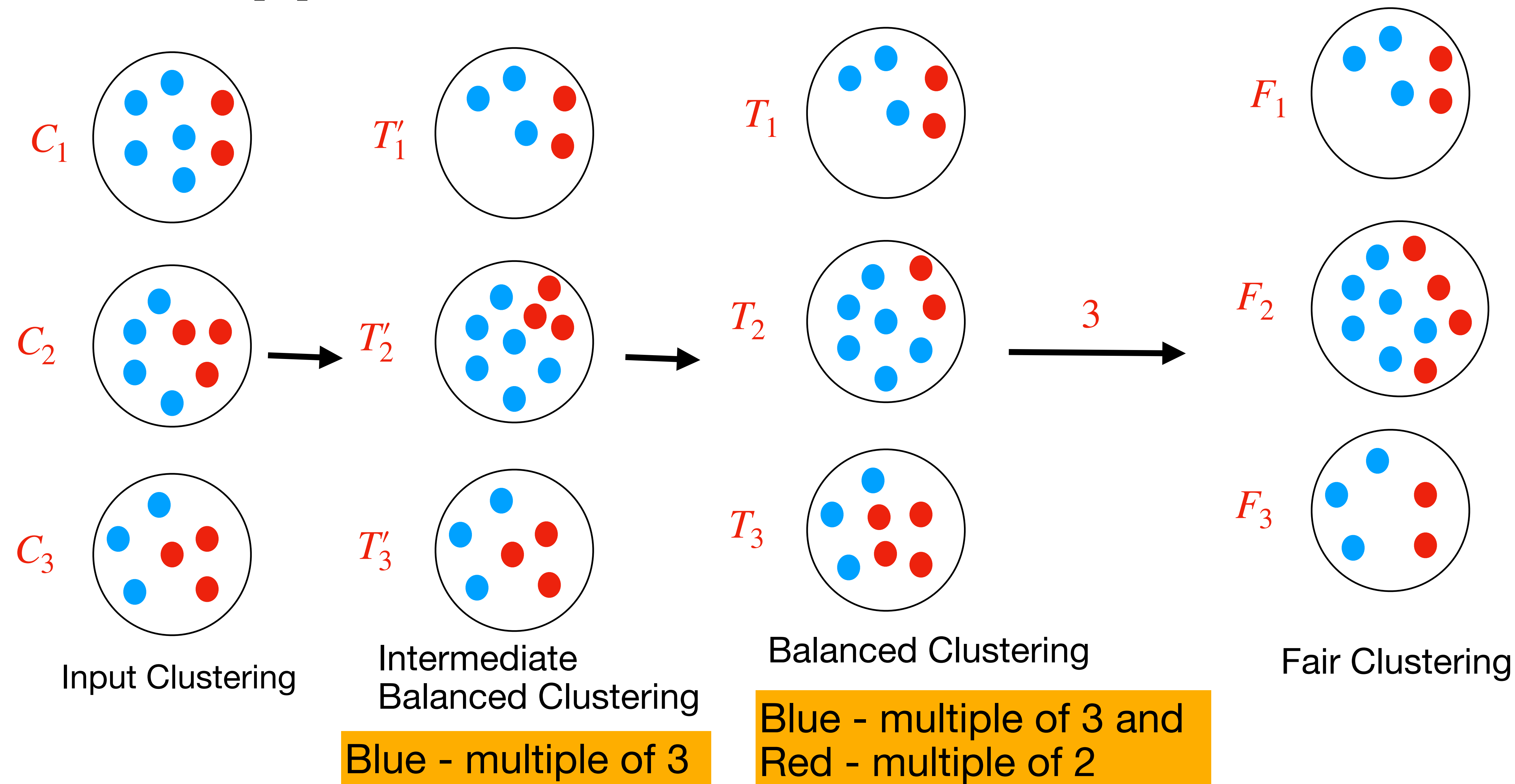


Approach for p:q case

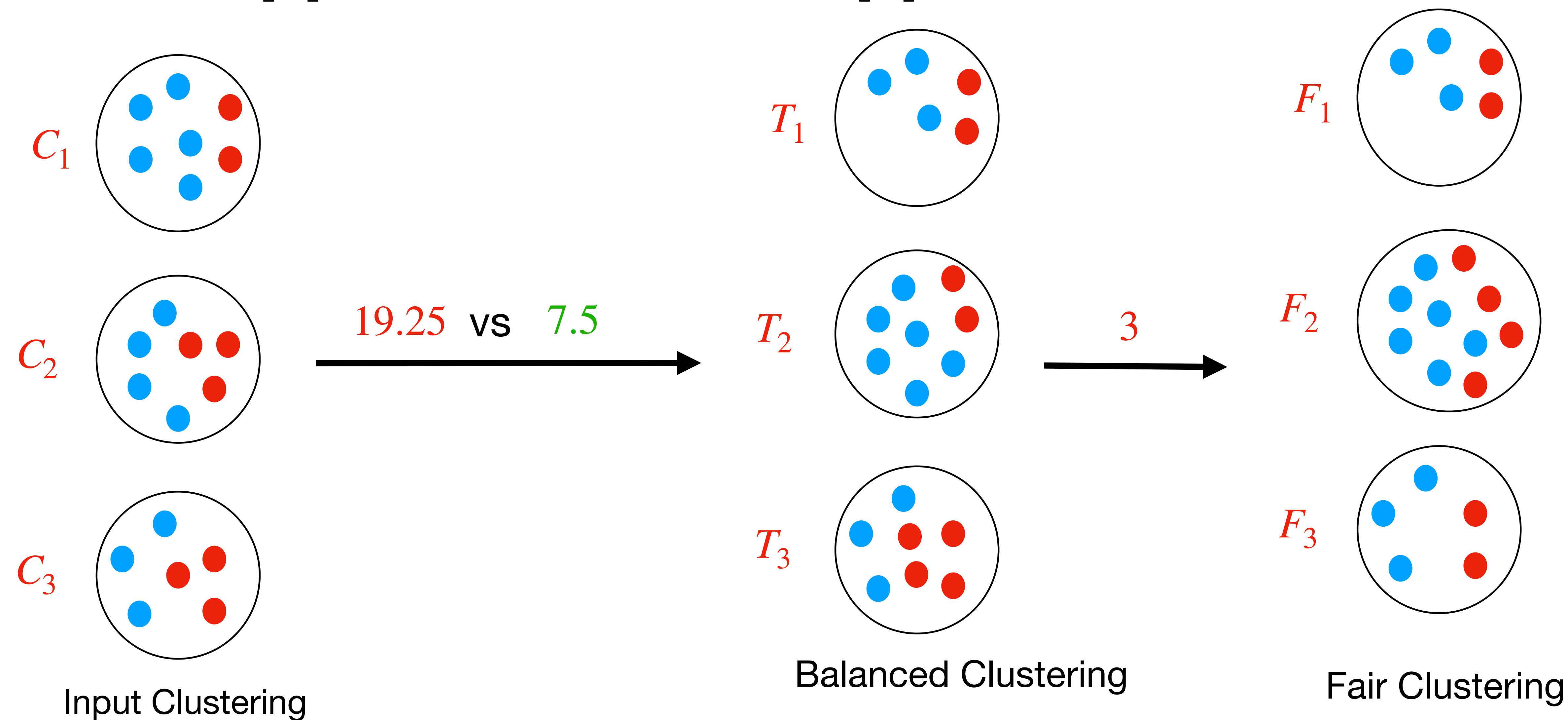


Overall approximation - 33 (independent of p,q)

Naive Approach



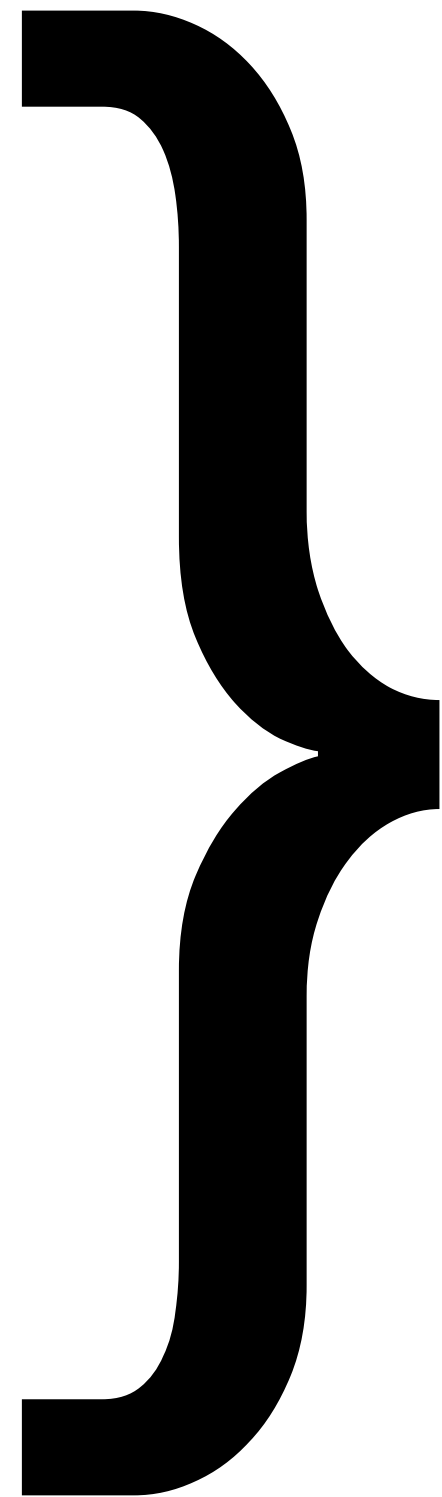
Naive Approach vs Our Approach



Overall approximation - 80 vs 33

Open Questions

- Multiple Colours.
- Overlapping subgroups.
- k-clusterings.
- Other notions of fairness.



Solve closest fair clustering problem for any of these settings.

Thank You For Your Attention